DBMS – SQL QUERY IMPLEMENTATION FOR E-COMMERCE DATABASE

StyleNova – Fashion Products E-Commerce Order Management System

Objective

1. Perform SELECT, WHERE, ORDER BY, and DISTINCT queries.

2. Search products based on price, category, and availability.

3. Retrieve customer and product information.

4. Apply filtering conditions.

5. Generate basic business reports.

1. Retrieve All Product Records

Display all fashion products available in the StyleNova database.

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
       102
Denim Jeans
         1       1299         15

       103
Floral Kurti
         2        899         20

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------

       104
Kids Hoodie
         3        749         12
```

2. Retrieve Selected Product Information

```
SQL> SELECT Product_ID, Product_Name, Category_ID, Price, Stock
  2  FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
       102
Denim Jeans
          1       1299         15

       103
Floral Kurti
          2        899         20

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------

       104
Kids Hoodie
          3        749         12

       106
Women Sneakers

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
          2       1199         18

       101
Cotton T-Shirt
          1        599         25
```

3. Apply WHERE Condition

Display products whose price is greater than ₹1,000.

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price > 1000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       102
Denim Jeans
      1299

       106
Women Sneakers
      1199

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
```

4. Search Products by Price Range

Display products priced between ₹500 and ₹1,500.

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price BETWEEN 500 AND 1500
  4  ORDER BY Price;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       101
Cotton T-Shirt
       599

       104
Kids Hoodie
       749

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       103
Floral Kurti
       899

       106
Women Sneakers

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1199

       102
Denim Jeans
      1299
```

5. Search Products by Category

```
SQL> SELECT p.Product_ID,
  2         p.Product_Name,
  3         c.Category_Name,
  4         p.Price
  5  FROM Product p
  6  JOIN Category c
  7  ON p.Category_ID = c.Category_ID
  8  WHERE c.Category_Name = 'Women';

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      PRICE
-------------------------------------------------- ----------
       103
Floral Kurti
Women                                                 899

       106
Women Sneakers
Women                                                1199

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      PRICE
-------------------------------------------------- ----------
```

6. Search Available Products

Display products currently in stock.

```
SQL> SELECT Product_ID, Product_Name, Stock
  2  FROM Product
  3  WHERE Stock > 0
  4  ORDER BY Product_Name;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     STOCK
----------
       101
Cotton T-Shirt
        25

       102
Denim Jeans
        15

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     STOCK
----------

       103
Floral Kurti
        20

       104
Kids Hoodie

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     STOCK
----------
        12

       106
Women Sneakers
        18
```

7. Search Unavailable Products

Display products whose stock quantity is zero.

```
SQL> SELECT Product_ID, Product_Name, Stock
  2  FROM Product
  3  WHERE Stock = 0;

no rows selected
```

8. Use DISTINCT

Display each category only once.

```
SQL> SELECT DISTINCT Category_ID
  2  FROM Product
  3  ORDER BY Category_ID;

CATEGORY_ID
-----------
          1
          2
          3
```

9. Sort Products Using ORDER BY

Display products from lowest price to highest price.

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       101
Cotton T-Shirt
       599

       104
Kids Hoodie
       749

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       103
Floral Kurti
       899

       106
Women Sneakers

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1199

       102
Denim Jeans
      1299
```

Conclusion

The StyleNova SQL Query Implementation task demonstrates how SELECT, WHERE, ORDER BY, and DISTINCT retrieve and filter information from an e-commerce database. Price, category, and stock conditions help search fashion products, while customer queries and summary reports support basic business analysis.